# Cessna 210R Centurion (C210R)

Continental IO-520-L, 300 hp · 6 seats · Typical 210R (1985–86, IO-520-L, 4,100 lb takeoff / 3,800 lb landing, 38 ft 10 in wing, unpressurised) POH procedures: electric hydraulic power pack with NO main gear doors (wheels retract into open wells and are held up by hydraulic pressure), hand-pump emergency extension, electric pre-select flaps, LEFT/RIGHT/OFF fuel selector with vapour return to the selected tank; the door-equipped 210M and the turbocharged T210R / P210R differ.

## V-speeds & limits

| | |
|---|---|
| Vr | **65–70 KIAS (liftoff, flaps 0–10°)** |
| Vx | **72 KIAS (sea level, flaps up)** |
| Vy | **95 KIAS (sea level, flaps up)** |
| Best glide | **85–90 KIAS (4,100 lb, gear & flaps up, prop low RPM)** |
| Vfe | **160 KIAS (10°) / 140 KIAS (20°) / 120–140 KIAS (30° FULL – verify placard; 210R checklists differ)** |
| Vlo | **140 KIAS (extend & retract)** |
| Vle | **165 KIAS (extended; verify placard)** |
| Va | **approx 133 KIAS at 4,100 lb (lower when lighter, approx 120 KIAS at 3,300 lb)** |
| Vno | **167 KIAS** |
| Vne | **204 KIAS** |
| Vs / Vso | **approx 66 / 59 KIAS (4,100 lb)** |
| Vref | **75–80 KIAS final (flaps 30°); 70–75 KIAS short field; 85–90 KIAS flaps up** |

## Normal procedures

### Before Start

| | |
|---|---|
| Preflight inspection | **COMPLETE (fuel sumps drained, oil quantity, open main gear wells clear – no gear doors on the R model, hydraulic fluid level checked)** |
| Passenger briefing | **COMPLETE (belts, both cabin doors, exits, no smoking)** |
| Seats, belts, shoulder harnesses | **ADJUST & LOCK** |
| Cabin doors (both) / baggage door | **CLOSED & LATCHED** |
| Parking brake | **SET** |
| Avionics power switch | **OFF** |
| Electrical equipment | **OFF** |
| Circuit breakers | **CHECK IN (including landing gear pump)** |
| Landing gear lever | **DOWN** |
| Fuel selector | **LEFT or RIGHT – fullest tank (there is no BOTH; feel for the detent)** |
| Fuel quantity | **CHECK both gauges against visual check (approx 87–90 gal usable total – verify POH)** |
| Cowl flaps | **OPEN** |
| Alternate static source | **OFF (normal)** |
| Master switch (BAT & ALT) | **ON** |
| Gear-down light (green) | **ON (press to test gear-up light)** |
| Low-voltage light | **OUT after start (may be on with engine stopped)** |
| Beacon | **ON** |

## Engine Start

| | |
|---|---|
| Propeller area | **CLEAR** |
| Propeller | **HIGH RPM (full forward)** |
| Throttle | **OPEN approx 1/4 to 1/2 in** |
| Mixture | **IDLE CUTOFF** |

**▶ Engine hot or vapour-locked?**

**No – normal prime:**

| | |
|---|---|
| Auxiliary fuel pump | **ON** |
| Mixture | **FULL RICH until fuel flow indicates approx 6–8 GPH (3–5 sec), then IDLE CUTOFF** |
| Auxiliary fuel pump | **OFF** |

**Yes – hot start:**

| | |
|---|---|
| Throttle | **FULL OPEN** |
| Auxiliary fuel pump | **ON approx 20–30 sec with mixture IDLE CUTOFF to purge vapour from the lines, then OFF** |
| Throttle | **1/4 to 1/2 in OPEN** |
| Mixture | **FULL RICH for approx 2 sec, then IDLE CUTOFF** |

If the engine floods: mixture IDLE CUTOFF, throttle FULL OPEN, crank; retard throttle and advance mixture as it fires.

| | |
|---|---|
| Ignition switch | **START (release when engine fires)** |
| Mixture | **ADVANCE SMOOTHLY to FULL RICH as engine fires** |
| Throttle | **800–1,000 RPM** |
| Oil pressure | **10 psi minimum within 30 sec (normal 30–60 psi)** |
| Ammeter / low-voltage light | **CHARGING / OUT** |
| Avionics power switch | **ON** |
| Hydraulic pump | **NOTE – may run briefly to build pressure, must stop** |
| Engine instruments | **CHECK** |

## Before Taxi

| | |
|---|---|
| Brakes | **CHECK on initial roll** |
| Nosewheel steering | **CHECK** |
| Flight instruments | **CHECK & SET (altimeter, DG / HSI, turn coordinator and compass during turns)** |
| Avionics / GPS / radios | **SET** |
| Autopilot / electric trim | **CHECK & OFF** |
| Elevator & rudder trim | **SET for takeoff** |
| Fuel selector | **CHECK fullest tank (LEFT or RIGHT)** |
| Cowl flaps | **OPEN** |

### Before Takeoff

| | |
|---|---|
| Parking brake | **SET** |
| Cabin doors / windows | **CLOSED & LATCHED** |
| Flight controls | **FREE & CORRECT** |
| Mixture | **FULL RICH (lean for smooth run-up above approx 3,000 ft elevation)** |
| Throttle | **1,700 RPM** |
| Magnetos | **CHECK L & R (150 RPM max drop, 50 RPM max difference)** |
| Propeller | **CYCLE 2–3 times (RPM drops approx 300–500, then returns), then HIGH RPM** |
| Engine instruments | **CHECK (oil pressure 30–60 psi, oil temp in green, fuel flow steady)** |
| Ammeter / low-voltage light | **CHARGING / OUT** |
| Suction | **4.5–5.5 in Hg** |
| Throttle | **IDLE check (approx 600–650 RPM), then 1,000 RPM** |
| Fuel selector | **FULLEST TANK (LEFT or RIGHT – feel for detent)** |
| Auxiliary fuel pump | **OFF** |
| Cowl flaps | **OPEN** |
| Propeller | **HIGH RPM** |
| Flaps | **0°–10° (pre-select lever, check indicator; 10° for short field)** |
| Trim | **SET takeoff** |
| Landing gear lever | **DOWN – green light ON** |
| Pitot heat | **AS REQUIRED** |
| Lights | **AS REQUIRED** |
| Transponder | **ALT** |
| Takeoff briefing | **COMPLETE (full throttle 2,850 RPM; 4,100 lb max takeoff, 3,800 lb max landing weight; abort point; gear up only with positive rate and no usable runway)** |

### After Takeoff / Climb

| | |
|---|---|
| Brakes | **APPLY momentarily to stop wheel rotation** |
| Landing gear | **UP (positive rate, no usable runway, below 140 KIAS)** |
| Gear-up light | **ON, hydraulic pump stopped (no main gear doors – wheels sit in the open wells, held up by pressure)** |
| Flaps | **UP (above 70–75 KIAS, after obstacles)** |
| Climb power | **25 in Hg / 2,550 RPM cruise climb (full throttle / 2,700 RPM max continuous; 2,850 RPM takeoff limited to 5 min)** |
| Mixture | **LEAN per POH climb fuel-flow schedule above approx 5,000 ft** |
| Cowl flaps | **OPEN (CHT 460°F max, oil temp 240°F max)** |
| Auxiliary fuel pump | **OFF (ON briefly if fuel flow fluctuates)** |
| Airspeed | **110–120 KIAS cruise climb (95 KIAS Vy)** |
| Engine instruments | **CHECK** |

### Cruise

| | |
|---|---|
| Cruise power | **SET (2,300–2,500 RPM, 20–25 in Hg typical, 75% max)** |
| Mixture | **LEAN per POH cruise table / EGT (approx 50°F rich of peak)** |
| Cowl flaps | **CLOSED (as required for CHT)** |
| Fuel selector | **ALTERNATE LEFT / RIGHT every 30–60 min – note the time, keep tanks within approx 10 gal** |
| Fuel quantity / flow | **CHECK against plan (vapour and excess fuel return to the SELECTED tank – do not run a tank to empty)** |
| Engine instruments | **CHECK (oil pressure 30–60 psi, oil temp 240°F max)** |
| Hydraulic pump | **BRIEF RUNS NORMAL (gear is held up by hydraulic pressure, no uplocks); continuous running is abnormal** |
| Altimeter / heading | **CHECK** |

### Descent

| | |
|---|---|
| Altimeter | **SET** |
| Fuel selector | **FULLEST TANK (LEFT or RIGHT)** |
| Mixture | **ENRICH progressively** |
| Power | **REDUCE gradually (avoid rapid cooling, keep CHT in green)** |
| Cowl flaps | **CLOSED** |
| Airspeed | **PLAN below 140 KIAS before gear extension** |
| Approach briefing | **COMPLETE** |
| Seat belts / harnesses | **SECURE** |

### Before Landing

| | |
|---|---|
| Fuel selector | **FULLEST TANK (LEFT or RIGHT)** |
| Auxiliary fuel pump | **OFF** |
| Mixture | **FULL RICH** |
| Cowl flaps | **AS REQUIRED** |
| Airspeed | **BELOW 140 KIAS** |
| Landing gear lever | **DOWN – green light ON, pump stopped (no door cycle to hear on the R model – allow approx 10 sec)** |
| Flaps | **10° below 160 KIAS; 20° below 140 KIAS; FULL (30°) on final below 120–140 KIAS (per placard)** |
| Propeller | **HIGH RPM (on final)** |
| Airspeed | **75–80 KIAS final (70–75 KIAS short field)** |
| Landing light | **ON** |
| Landing gear | **GREEN – CONFIRMED on final, horn silent at idle** |

### After Landing

| | |
|---|---|
| Flaps | **UP** |
| Cowl flaps | **OPEN** |
| Auxiliary fuel pump | **OFF** |
| Pitot heat | **OFF** |
| Landing light / strobes | **AS REQUIRED** |
| Transponder | **STBY / as required** |
| Trim | **SET takeoff** |

### Shutdown

| | |
|---|---|
| Parking brake | **SET** |
| Throttle | **1,000 RPM (cool-down approx 1 min)** |
| Avionics power switch | **OFF** |
| Electrical equipment / lights | **OFF** |
| Mixture | **IDLE CUTOFF** |
| Ignition switch | **OFF (after propeller stops), key removed** |
| Master switch (BAT & ALT) | **OFF** |
| Fuel selector | **LEFT or RIGHT (leave on a tank, not OFF, to avoid cross-feed confusion; follow operator policy)** |
| Control lock | **INSTALL** |
| Chocks / tiedowns | **AS REQUIRED** |

## Abnormal procedures

### Alternator Failure / Low Voltage

Low-voltage light on, ammeter discharging (28 V system)

Avionics power switch ..... **OFF**

Alternator circuit breaker ..... **CHECK IN / RESET once**

Master switch ..... **OFF, then ON (both halves)**

**▶ Low-voltage light out, ammeter charging?**

**Yes:**

- Avionics power switch ..... **ON**
- Ammeter ..... **MONITOR**

**No:**

- Master switch ..... **ALT half OFF (BAT ON)**
- Non-essential electrical load ..... **REDUCE (battery only – pitot heat, lights, extra radios OFF)**
- Landing gear ..... **PLAN early extension while battery power remains (hand pump if the pump will not run)**
- Flaps ..... **PLAN to extend early or land flaps-up**
- Deferred: Alternator failed – battery only; extend gear early (hand pump if needed), land as soon as practical. ..... **NOTED**

### Low Oil Pressure / High Oil Temperature

Oil pressure below 30 psi in cruise or oil temperature above 240°F

Oil pressure / oil temperature ..... **CHECK and cross-check**

Cowl flaps ..... **OPEN**

Mixture ..... **ENRICH**

**▶ Oil temperature rising or pressure below 10 psi?**

**Yes – genuine:**

- Power ..... **REDUCE to minimum required**
- Airport ..... **DIVERT to nearest suitable**
- Engine ..... **EXPECT FAILURE – stay within glide of a landing area, plan gear extension early**
- Deferred: Low oil pressure / high oil temperature – land as soon as possible, expect possible engine failure. ..... **NOTED**

**No – gauge suspect:**

- Flight ..... **CONTINUE, monitor oil temperature and pressure**
- Deferred: Oil pressure / temperature indication suspect – maintenance check required. ..... **NOTED**

### Fuel Flow Fluctuation / Vapour / Rough Running

Auxiliary fuel pump ..... **ON until fuel flow is steady (approx 1–2 min), then OFF**

Fuel selector ..... **SWITCH TANKS (fullest tank; feel for detent)**

Mixture ..... **ADJUST for smooth running**

Magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (fuel flow, CHT, EGT, fuel quantity on selected tank)**

**▶ Engine running smoothly, fuel flow steady?**

**Yes:**

- Deferred: Fuel flow / rough running cleared – leave aux pump OFF unless needed again; report to maintenance. ..... **NOTED**

**No:**

- Auxiliary fuel pump ..... **ON (leave on)**
- Power ..... **REDUCE to smoothest setting**
- Deferred: Engine rough / fuel flow unsteady – land as soon as practical at nearest suitable airport. ..... **NOTED**

## Hydraulic Pump Runs Continuously

Pump heard running more than approx 1 min after a gear cycle, or running in cruise

Landing gear lever ..... **CHECK in the detent for the current gear position**

Gear lights ..... **CHECK (gear up with no gear-up light = gear not fully up)**

**▶ Pump still running?**

**Yes:**

Landing gear pump circuit breaker ..... **PULL (prevents pump burnout; hydraulic leak or pressure switch fault)**

Without pump pressure the gear is not locked up: expect the gear to sag out of the open wells and airspeed / performance to suffer. Keep below 140 KIAS.

Airspeed ..... **BELOW 140 KIAS**

Landing gear ..... **EXTEND with the hand pump (see Emergency Gear Extension) before landing**

Deferred: Hydraulic pump isolated – gear extended by hand pump, do not retract; hydraulic system inspection required. ..... **NOTED**

**No:**

Deferred: Hydraulic pump cycled abnormally – note for maintenance; check gear lights before landing. ..... **NOTED**

## Gear Unsafe Indication / Gear Warning Horn

No gear-up or gear-down light after cycle, or horn sounds unexpectedly

Throttle ..... **NOTE – horn sounds when throttle is retarded below approx 12 in Hg with the gear not down and locked (also with flaps beyond approx 20° and gear up on some aircraft)**

Airspeed ..... **BELOW 140 KIAS**

Landing gear lever ..... **CHECK full travel into detent**

Landing gear pump circuit breaker ..... **CHECK IN / RESET once**

Gear lights ..... **PRESS TO TEST / swap bulbs if possible**

Landing gear lever ..... **RECYCLE once to the desired position (allow approx 10 sec; listen for the pump to stop)**

**▶ Gear in desired position with correct light?**

**Yes:**

Gear indication ..... **CONFIRM; horn silent at idle with gear down**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

**No – will not retract:**

Landing gear lever ..... **DOWN – leave down, green light**

Airspeed ..... **BELOW 140 KIAS**

Deferred: Gear will not retract – gear down, reduced performance and range, land as soon as practical. ..... **NOTED**

**No – will not extend:**

Use Emergency Gear Extension (Hand Pump).

## Flap Failure / Asymmetry

Flap lever ..... **RETURN to the current position – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

**▶ Flaps operable and symmetric?**

**Yes:**

Flap setting ..... **AS REQUIRED (stay below 160 / 140 KIAS limits)**

**No:**

Landing ..... **AT CURRENT FLAP SETTING – flaps up approx 85–90 KIAS final, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance. ..... **NOTED**

## Propeller Overspeed

RPM above 2,850 and not responding to prop control

Throttle ..... **REDUCE**

Propeller ..... **LOW RPM (full aft)**

Airspeed ..... **REDUCE (pitch up) to bring RPM within limits**

Oil pressure ..... **CHECK (governor is oil-operated)**

Power ..... **SET minimum that keeps RPM at or below 2,700**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

### Cabin or Baggage Door Open in Flight

| | |
|---|---|
| Aircraft | **CONTROLLABLE – fly the airplane** |
| Airspeed | **REDUCE to approx 100 KIAS** |
| Belts / loose items | **SECURE** |
| Cabin door | **OPEN STORM WINDOW / vents, then pull door closed and latch if it can be reached safely (co-pilot side only)** |

Do not let a door distract from flying. A door that will not latch stays slightly open; expect noise and some buffet.

| | |
|---|---|
| Deferred: Door unlatched – return and land as soon as practical, normal approach speed. | **NOTED** |

### Vacuum / Attitude Indicator Failure

Suction below 4.5 in Hg or gyro flags

| | |
|---|---|
| Suction gauge | **CHECK** |
| Attitude / heading indicators | **CROSS-CHECK with turn coordinator, compass, GPS, standby instruments** |
| Failed instruments | **COVER** |
| Autopilot | **OFF if coupled to the failed gyro** |
| IMC | **EXIT / AVOID** |
| Deferred: Vacuum / gyro failure – partial panel; prefer VMC approach. | **NOTED** |

### Inadvertent Icing Encounter

| | |
|---|---|
| Pitot heat | **ON** |
| Heading / altitude | **TURN BACK or CHANGE ALTITUDE to exit icing conditions** |
| Cabin heat / defroster | **FULL ON (windshield)** |
| Power | **INCREASE to minimise ice build-up on the propeller and to maintain airspeed; cycle RPM to shed prop ice** |
| Cowl flaps | **ADJUST to keep CHT in green (induction icing: expect power loss)** |
| Alternate static source | **ON if static ports suspect** |

Stall speed rises with ice. Avoid steep turns; use flaps 0°–10° only and approach at 90–100 KIAS. Do not retract flaps once lowered if ice is on the tail.

| | |
|---|---|
| Landing gear | **EXTEND EARLY (below 140 KIAS) – ice in the open gear wells may slow extension** |
| Deferred: Ice on airframe – flaps 0°–10°, approach 90–100 KIAS, longer landing distance, higher stall speed. | **NOTED** |

### Pitot / Static Source Blocked

Airspeed, altimeter or VSI erratic

| | |
|---|---|
| Pitot heat | **ON** |
| Alternate static source | **ON (pull) – expect altimeter and airspeed to read differently; see POH calibration** |
| Cabin vents / windows | **CLOSED when using alternate static** |
| Attitude / power | **FLY KNOWN PITCH AND POWER (GPS ground speed as cross-check)** |
| Deferred: Pitot / static unreliable – fly known pitch and power on approach, add margin to approach speed. | **NOTED** |

## Emergency procedures

### Engine Failure During Takeoff Roll

| | |
|---|---|
| Throttle (memory) | **CLOSED** |
| Brakes (memory) | **AS REQUIRED – stop straight ahead** |
| Flaps | **UP** |
| Mixture | **IDLE CUTOFF** |
| Ignition switch | **OFF** |
| Fuel selector | **OFF** |
| Master switch | **OFF** |

## Engine Failure After Takeoff

Airborne at low altitude

**LAND IMMEDIATELY**

| | |
|---|---|
| Airspeed (memory) | **75–80 KIAS flaps down / 85 KIAS flaps up** |

**▶ Runway remaining to land on?**

**Yes:**

| | |
|---|---|
| Landing gear | **DOWN (if retracted and time permits – approx 10 sec cycle, no doors to wait for)** |
| Flaps | **AS REQUIRED** |
| Land | **STRAIGHT AHEAD** |

**No – land ahead off-field:**

| | |
|---|---|
| Landing gear | **DOWN on firm surface / UP on rough terrain or water** |
| Flaps | **30° when landing assured** |
| Mixture | **IDLE CUTOFF** |
| Fuel selector | **OFF** |
| Ignition switch | **OFF** |
| Master switch | **OFF (after gear and flaps set – both are electric)** |

Land straight ahead with only small changes of direction. Do not attempt a turn back.

## Engine Failure in Flight

Loss of power with altitude available

**LAND AS SOON AS POSSIBLE**

- Airspeed (memory) — **85 KIAS (best glide, gear & flaps up)**
- Propeller (memory) — **LOW RPM (full aft) to extend glide**
- Fuel selector (memory) — **SWITCH TANKS (feel for detent)**
- Auxiliary fuel pump (memory) — **ON (check fuel flow)**
- Mixture (memory) — **FULL RICH**
- Ignition switch (memory) — **BOTH (check)**

**▶ Restart attempt practicable?**

**Yes – restart:**

- Throttle — **1/2 OPEN**
- Starter — **ENGAGE if propeller has stopped**

**▶ Engine restarted?**

**Yes:**

- Power — **AS REQUIRED**
- Auxiliary fuel pump — **OFF once fuel flow is steady (ON if it fluctuates)**
- Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel system or vapour. — **NOTED**

**No:**

- Auxiliary fuel pump — **OFF**

Continue with forced landing.

**No – forced landing:**

Forced landing without power:

- Seats / belts / harnesses — **SECURE**
- Field — **SELECT – plan to arrive high, flaps and gear only when assured**

**▶ Surface firm and long enough for a gear-down landing?**

**Yes – firm surface:**

- Landing gear — **DOWN below 140 KIAS (hand pump if hydraulic pump fails – allow time; no doors to open)**
- Gear-down light — **GREEN**

**No – rough terrain, soft ground or water:**

- Landing gear — **UP (leave retracted in the open wells)**

Gear-up touchdown: minimum speed, slightly nose high, wings level. Unlatch doors before touchdown.

- Flaps — **30° when landing assured**
- Airspeed — **75–80 KIAS on final (85 KIAS flaps up)**
- Mixture — **IDLE CUTOFF**
- Fuel selector — **OFF**
- Ignition switch — **OFF**
- Master switch — **OFF (after gear and flaps set)**
- Cabin doors — **UNLATCH before touchdown**

## Engine Fire in Flight

**LAND IMMEDIATELY**

- Mixture (memory) — **IDLE CUTOFF**
- Fuel selector (memory) — **OFF**
- Auxiliary fuel pump (memory) — **OFF**
- Landing gear — **DOWN now if a gear-down landing is planned (below 140 KIAS – electric pump needs the master ON)**
- Master switch — **OFF**
- Cabin heat / cabin air — **OFF (overhead vents open as required)**
- Airspeed — **100 KIAS – if fire persists, increase speed to find one that gives an incombustible mixture**
- Forced landing — **EXECUTE (do not attempt restart)**

With the master OFF the gear can only be extended with the hand pump and the flaps will not move.

## Engine Fire During Start

Starter (memory) ..... **CONTINUE CRANKING to draw fire into engine**

▶ **Engine starts?**

**Yes:**

- Throttle ..... **1,700 RPM for a few minutes**
- Engine ..... **SHUT DOWN and inspect for damage**

**No:**

- Mixture ..... **IDLE CUTOFF**
- Throttle ..... **FULL OPEN**
- Auxiliary fuel pump ..... **OFF**
- Fuel selector ..... **OFF**
- Starter ..... **CONTINUE CRANKING**
- Ignition switch ..... **OFF**
- Master switch ..... **OFF**
- Fire extinguisher ..... **OBTAIN / USE**
- Aircraft ..... **EVACUATE**

## Electrical / Cabin Fire

**LAND AS SOON AS POSSIBLE**

Master switch (memory) ..... **OFF**

Avionics power switch (memory) ..... **OFF**

All other switches (except ignition) (memory) ..... **OFF**

Cabin heat / vents ..... **CLOSED**

Fire extinguisher ..... **ACTIVATE (then ventilate cabin once fire is out)**

▶ **Fire out and electrical power required?**

**Yes:**

- Circuit breakers ..... **CHECK – do not reset a faulted breaker**
- Master switch ..... **ON**
- Avionics power switch / electrical equipment ..... **ON one at a time, pausing to check for smoke**
- Landing gear ..... **EXTEND EARLY while power is available**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

**No:**

Gear pump and flaps are electric: plan a hand-pump gear extension (see Emergency Gear Extension) and a flaps-up landing at 85–90 KIAS.

- Deferred: Electrical power off – hand-pump gear extension, flaps-up landing, no landing light or radio. ..... **NOTED**

## Emergency Descent

Throttle (memory) ..... **CLOSED**

Propeller (memory) ..... **HIGH RPM**

Landing gear (memory) ..... **DOWN (below 140 KIAS)**

Flaps (memory) ..... **UP**

Airspeed (memory) ..... **165 KIAS gear down (Vle); Va approx 133 KIAS in turbulence**

Mixture ..... **ENRICH as required**

Cowl flaps ..... **CLOSED**

## Emergency Gear Extension (Hand Pump)

Gear does not extend hydraulically or no green light

| | |
|---|---|
| Airspeed | **BELOW 140 KIAS (slower reduces pump effort – approx 100 KIAS)** |
| Master switch | **ON** |
| Landing gear lever | **DOWN** |
| Landing gear pump circuit breaker | **PULL if the pump runs continuously or has tripped (otherwise leave in)** |
| Emergency hand pump | **EXTEND HANDLE (floor between front seats) and PUMP until the green light comes on and the handle becomes firm (no main gear doors – gear extends straight into the airstream; typically 40–60 strokes)** |
| Gear-down light | **GREEN** |
| Hand pump handle | **STOW** |

Do not retract the gear or reset the pump breaker in flight after a hand-pump extension. Leave the gear down.

**▶ Green light obtained?**

**Yes:**

Deferred: Gear extended with the hand pump – do not retract; gear and hydraulic system inspection required. ..... **NOTED**

**No:**

Gear-down light bulb ..... **CHECK (swap with gear-up bulb if possible); gear horn silent at idle?**

If still unsafe, continue to Gear-Up / Partial Gear Landing procedure.

## Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

| | |
|---|---|
| Runway | **LONGEST HARD SURFACE available, emergency services advised** |
| Passenger briefing | **COMPLETE (belts tight, brace, exits, both doors)** |

**▶ Gear position?**

**All up:**

Landing gear lever ..... **UP – do not cycle**

**Nose gear only down:**

Landing gear ..... **LEAVE AS IS – land on nose gear and belly, hold nose up as long as practical**

**One main gear up:**

Landing gear ..... **RETRACT ALL if possible (landing on one main is worse than gear up); otherwise land on the extended gear side of the runway, wing low to that side**

| | |
|---|---|
| Flaps | **30°** |
| Approach | **NORMAL, 75–80 KIAS** |
| Auxiliary fuel pump | **OFF** |
| Cabin doors | **UNLATCHED** |

On short final:

| | |
|---|---|
| Mixture | **IDLE CUTOFF** |
| Fuel selector | **OFF** |
| Ignition switch | **OFF** |
| Master switch | **OFF** |
| Touchdown | **MINIMUM SPEED, slightly nose high, wings level** |
| Aircraft | **EVACUATE once stopped** |

## Inadvertent Spin

| | |
|---|---|
| Throttle (memory) | **IDLE** |
| Ailerons (memory) | **NEUTRAL** |
| Rudder (memory) | **FULL OPPOSITE to rotation** |
| Elevator (memory) | **BRISKLY FORWARD to break stall (just past neutral)** |
| Flaps | **UP (if extended)** |
| Rudder | **NEUTRAL when rotation stops** |
| Recovery | **SMOOTH PULL-OUT from the dive – respect Vne 204 KIAS and gear / flap speeds** |

Intentional spins are prohibited. Lower nose first if a secondary stall threatens.